In 2015, the restaurant processed 21,350 orders generating approximately $817K in total revenue, with an average order value of $38.31. Demand peaks sharply at lunch (12–1 PM) and dinner (6–7 PM), with Fridays being the busiest day. Classic and Supreme pizzas lead revenue contribution, while chicken-based pizzas dominate the top revenue earners. The restaurant's seating capacity appears adequate for current demand, though kitchen pressure during peak hours warrants staffing attention. Key opportunities lie in bundle promotions, off-peak incentives, and reviewing underperforming menu items like the Brie Carre Pizza.

```sql
-- ============================================================
-- SECTION 1: DATA VALIDATION
-- ============================================================

-- 1.1 Row counts for each table
SELECT 'pizza_types'   AS table_name, COUNT(*) AS row_count FROM pizza_types
UNION ALL
SELECT 'pizzas',        COUNT(*) FROM pizzas
UNION ALL
SELECT 'orders',        COUNT(*) FROM orders
UNION ALL
SELECT 'order_details', COUNT(*) FROM order_details;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

table_name	row_count
pizza_types	32
pizzas	96
orders	21350
order_details	48620

Business Insight

Before analyzing restaurant performance, the dataset was validated to ensure the information is complete and reliable. The data contains 32 pizza types, 96 pizza variations, 21,350 total customer orders, and 48,620 individual order line items. Additional checks confirmed that there are no missing values in critical fields such as order ID, date, time, pizza ID, or quantity. The validation also shows that all order quantities are positive and every order detail correctly corresponds to a valid order. The dataset spans the full calendar year from January 1, 2015 to December 31, 2015, providing a complete year of sales activity for analysis.

Ensuring data quality before performing business analysis is essential for producing accurate insights and reliable conclusions. Since the dataset shows no missing records, invalid quantities, or broken order relationships, the restaurant’s transaction data can be considered consistent

and trustworthy. This provides a strong foundation for further analysis of customer demand patterns, product performance, and operational workload throughout the year.

```sql
-- 1.2 Check for NULL values in critical columns
SELECT
    SUM(order_id   IS NULL) AS null_order_id,
    SUM(date       IS NULL) AS null_date,
    SUM(time       IS NULL) AS null_time
FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

null_order_id	null_date	null_time
0	0	0

```sql
SELECT
    SUM(order_details_id IS NULL) AS null_detail_id,
    SUM(order_id         IS NULL) AS null_order_id,
    SUM(pizza_id         IS NULL) AS null_pizza_id,
    SUM(quantity         IS NULL) AS null_quantity
FROM order_details;

-- 1.3 Check for invalid quantities (should be > 0)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

null_detail_id	null_order_id	null_pizza_id	null_quantity
0	0	0	0

```sql
-- 1.3 Check for invalid quantities (should be > 0)
SELECT COUNT(*) AS invalid_quantity_rows
FROM order_details
WHERE quantity <= 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

invalid_quantity_rows
0

```sql
-- 1.4 Check for orphan order_ids in order_details
SELECT COUNT(*) AS orphan_order_details
FROM order_details od
LEFT JOIN orders o ON od.order_id = o.order_id
WHERE o.order_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orphan_order_details
0

```sql
-- 1.5 Date range of the dataset
SELECT MIN(date) AS start_date, MAX(date) AS end_date FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

start_date	end_date
2015-01-01	2015-12-31

What days and times is the restaurant busiest?

Busiest Hour

```sql
SELECT
HOUR(time) AS hour_of_day,
COUNT(order_id) AS total_orders
FROM orders
GROUP BY hour_of_day
ORDER BY total_orders DESC;
```

Business Insight

Customer orders are heavily concentrated around midday and early evening hours. The highest number of orders occurs at 12 PM with 2,520 orders, followed closely by 1 PM with 2,455 orders. A second surge appears in the evening around 6 PM and 7 PM, indicating strong dinner demand. These patterns suggest that customers primarily order pizza during typical meal times, particularly lunch breaks and evening dining hours. Order activity drops significantly late at night and during early morning hours.

hour_of_day	total_orders
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642
14	1472
15	1468
11	1231
21	1198
22	663
23	28
10	8
9	1

Recommendation

The restaurant should ensure that kitchen staff and service teams are fully prepared during the lunch and early evening rush periods. Preparing ingredients in advance and assigning additional staff during these peak hours can help reduce waiting times and maintain service quality. Promotions targeted at off-peak hours could also help distribute demand more evenly throughout the day.

Busiest Day of the Week

```sql
SELECT
DAYNAME(date) AS day_of_week,
COUNT(order_id) AS total_orders
FROM orders
GROUP BY day_of_week
ORDER BY total_orders DESC;
```

Business Insight

Order volume varies across the week, with Friday generating the highest number of orders at 3,538. Demand remains strong toward the end of the workweek, with Thursday and Saturday also showing high customer activity. This trend suggests that customers are more likely to order pizza as the weekend approaches, possibly for social gatherings, casual dining, or relaxed meals after work. In contrast, Sunday records the lowest number of orders among all days.

day_of_week	total_orders
Friday	3538
Thursday	3239
Saturday	3158
Wednesday	3024
Tuesday	2973
Monday	2794
Sunday	2624

Recommendation

The restaurant should allocate more staff and kitchen resources toward the end of the week to manage the increased customer demand. Special weekend promotions or group meal deals could further capitalize on the higher traffic during these days. Planning inventory based on these patterns will also help ensure that popular ingredients remain available during peak demand periods.

How many pizzas are made during peak periods?

Pizzas Produced During Peak Periods

```sql
SELECT
HOUR(o.time) AS hour,
SUM(od.quantity) AS pizzas_made
FROM orders o
JOIN order_details od
ON o.order_id = od.order_id
GROUP BY hour
ORDER BY pizzas_made DESC;
```

Business Insight

The production data shows that the kitchen experiences its highest workload during the lunch and early evening hours. Around 12 PM alone, the kitchen prepares approximately 6,776 pizzas, followed closely by 1 PM with about 6,413 pizzas. Another significant surge occurs during the evening between 5 PM and 7 PM, where pizza production remains consistently high. These peaks align closely with typical meal times, indicating that customer demand is highly concentrated during these periods. Such demand spikes place considerable pressure on kitchen operations within short time windows.

hour	pizzas_made
12	6776
13	6413
18	5417
17	5211
19	4406
16	4239
14	3613
20	3534
15	3216
11	2728
21	2545
22	1386
23	68
10	18
9	4

Recommendation

To manage these peak production periods effectively, the restaurant should ensure adequate kitchen staffing and streamline food preparation workflows during lunch and evening rush hours. Preparing dough, toppings, and ingredients in advance can help maintain faster order fulfillment. Additionally, organizing kitchen roles efficiently during busy hours can reduce preparation delays and ensure consistent service quality.

What are the best and worst selling pizzas?

Best Selling Pizzas

```sql
SELECT
pt.name,
SUM(od.quantity) AS pizzas_sold
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY pizzas_sold DESC
LIMIT 10;
```

Best Selling Pizzas

Business Insight

The sales data shows that a small group of pizzas consistently drives the highest demand among customers. The Classic Deluxe Pizza leads with 2,453 pizzas sold, followed closely by Barbecue Chicken, Hawaiian, and Pepperoni pizzas. These items appear to be widely preferred, likely due to their familiar flavors and broad customer appeal. The relatively small difference in sales among the top pizzas indicates that several menu items perform strongly rather than relying on a single dominant product. This suggests a balanced demand across multiple popular pizza varieties.

name	pizzas_sold
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371
The California Chicken Pizza	2370
The Sicilian Pizza	1938
The Spicy Italian Pizza	1924
The Southwest Chicken Pizza	1917
The Big Meat Pizza	1914

Recommendation

The restaurant should ensure that ingredients for these high-demand pizzas are always well stocked to avoid shortages during busy hours. Featuring these pizzas prominently in menus, combo offers, or promotional campaigns could further increase sales. Highlighting these popular choices can also simplify decision-making for customers and improve overall order value.

Worst Selling Pizzas

```sql
SELECT
pt.name,
SUM(od.quantity) AS pizzas_sold
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY pizzas_sold ASC
LIMIT 10;
```

Business Insight

The analysis also reveals several pizzas with significantly lower sales compared to the top-performing items. The Brie Carre Pizza records the lowest demand with only 490 pizzas sold, followed by Mediterranean and Calabrese pizzas with fewer than 1,000 orders each. These lower sales volumes suggest that these items may not strongly resonate with customer preferences or may have limited visibility on the menu. Maintaining multiple low-performing items can increase inventory complexity without contributing significantly to revenue.

name	pizzas_sold
The Brie Carre Pizza	490
The Mediterranean Pizza	934
The Calabrese Pizza	937
The Spinach Supreme Pizza	950
The Soppressata Pizza	961
The Spinach Pesto Pizza	970
The Chicken Pesto Pizza	973
The Italian Vegetables Pizza	981
The Chicken Alfredo Pizza	987
The Green Garden Pizza	997

Recommendation

Management should review these underperforming pizzas to determine whether they should be improved, repositioned, or potentially removed from the menu. Introducing limited-time promotions or adjusting ingredients could help test whether customer interest can be increased. Simplifying the menu by focusing on stronger performers may also improve kitchen efficiency and reduce ingredient waste.

What is the average order value?

Average Order Value

```sql
SELECT
ROUND(
SUM(od.quantity * p.price) / COUNT(DISTINCT o.order_id),
2
) AS avg_order_value
FROM orders o
JOIN order_details od
ON o.order_id = od.order_id
JOIN pizzas p
ON od.pizza_id = p.pizza_id;
```

Business Insight

The analysis shows that the restaurant generates an average of approximately $38.31 per order. This indicates that customers typically purchase more than a single pizza or combine their orders with additional items such as sides or drinks. A healthy average order value suggests that many transactions involve group dining or multiple-item purchases rather than individual meals. This purchasing behavior provides a strong foundation for increasing total revenue through strategic upselling.

	avg_order_value
▶	38.31

Recommendation

The restaurant can increase the average order value by offering bundle deals, family meal packages, or discounts for ordering multiple pizzas. Encouraging customers to add beverages, desserts, or side dishes during checkout could also raise transaction value. Even small increases in average order value can significantly improve total annual revenue when applied across thousands of orders.

How well is the restaurant utilizing seating capacity?
Restaurant capacity:
15 tables
60 seats

average orders per day.
Seating Capacity Utilization

```
SELECT
ROUND(COUNT(order_id) / 365,2) AS avg_orders_per_day
FROM orders;
```

Business Insight

The restaurant records an average of approximately 58 orders per day while operating with 15 tables and 60 seats. This indicates that the restaurant experiences a steady flow of customers throughout the day. Given the available seating capacity, the restaurant appears capable of handling the current demand without consistently exceeding its physical limits. However, during peak hours such as lunch and dinner, seating availability may become temporarily constrained due to the concentration of orders during those periods. Overall, the seating capacity seems adequate for the restaurant's daily order volume.

avg_orders_per_day
58.49

Recommendation

The restaurant should focus on managing table turnover efficiently during peak hours to maximize the use of available seating. Encouraging takeout and delivery options during busy periods could help reduce pressure on dine-in seating. Monitoring peak-time occupancy patterns may also help management decide whether future expansion or layout optimization would improve customer flow.

Average Pizzas per Order

```
ROUND(SUM(quantity) / COUNT(DISTINCT order_id),2) AS avg_pizzas_per_order
FROM order_details;
```

Business Insight

On average, customers purchase about 2.32 pizzas per order. This indicates that many transactions involve multiple pizzas rather than single-item purchases. Such behavior suggests that customers frequently order for small groups, families, or shared meals rather than individual consumption. This purchasing pattern reflects the social nature of pizza dining and highlights the importance of group-oriented ordering behavior.

	avg_pizzas_per_order
▶	2.32

Recommendation

The restaurant can capitalize on this behavior by introducing bundle deals or discounts for ordering multiple pizzas. Family meal offers or group combo packages could further encourage larger orders and increase overall revenue. Marketing campaigns can also emphasize shared dining experiences to align with this observed customer behavior.

Average Pizzas Produced per Day

```
SELECT
ROUND(SUM(quantity) / 365,2) AS avg_pizzas_per_day
FROM order_details;
```

Business Insight

The restaurant produces an average of approximately 136 pizzas per day. This consistent level of daily production indicates a stable demand for the restaurant's menu items. When combined with the average number of orders per day, it reinforces the observation that most customers order more than one pizza per transaction. This production volume provides a useful benchmark for planning ingredient inventory and kitchen workload.

	avg_pizzas_per_day
▶	135.82

Recommendation

Management should plan ingredient procurement and kitchen staffing based on this average daily production level. Maintaining sufficient stock of frequently used ingredients will help ensure smooth operations during busy periods. Monitoring changes in daily production trends over time could also help the restaurant anticipate growth in demand and adjust operational capacity accordingly.

Revenue Contribution by Pizza Category

```sql
SELECT
pt.category,
ROUND(SUM(od.quantity * p.price),2) AS revenue,
ROUND(
SUM(od.quantity * p.price) /
(
SELECT SUM(od.quantity * p.price)
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
) * 100
,2) AS revenue_percentage
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category
ORDER BY revenue DESC;
```

Business Insight

The revenue distribution across pizza categories shows that the Classic category generates the highest revenue, contributing approximately $220,053, which accounts for about 26.9% of total sales. This is followed closely by the Supreme category with about 25.5% of revenue. Chicken and Veggie pizzas also contribute significantly, each generating nearly one quarter of the total revenue. The relatively balanced distribution across categories indicates that customer demand is spread across multiple flavor profiles rather than being dominated by a single category. This diversification helps stabilize overall revenue by reducing dependence on one specific product group.

category	revenue	revenue_percentage
Classic	220053.10	26.91
Supreme	208197.00	25.46
Chicken	195919.50	23.96
Veggie	193690.45	23.68

Recommendation

The restaurant should continue promoting these high-performing categories while ensuring ingredient availability for their most popular pizzas. Introducing seasonal or limited-edition variations within these categories could help maintain customer interest and increase repeat purchases. Marketing campaigns that highlight category diversity may also attract a wider range of customer preferences.

Monthly Revenue Trend (Seasonality)

```
SELECT
MONTHNAME(o.date) AS month,
ROUND(SUM(od.quantity * p.price),2) AS monthly_revenue
FROM orders o
JOIN order_details od
ON o.order_id = od.order_id
JOIN pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY MONTH(o.date), MONTHNAME(o.date)
ORDER BY MONTH(o.date);
```

Business Insight

The monthly revenue trend shows that sales remain relatively consistent throughout the year, with moderate fluctuations across months. Revenue peaks around July at approximately $72,558, while September and October record the lowest monthly revenues, each slightly above $64,000. The variation between the highest and lowest months is not extremely large, suggesting stable demand for pizza throughout the year. However, slight increases during certain months may reflect seasonal dining behavior, local events, or holiday periods.

month	monthly_revenue
January	69793.30
February	65159.60
March	70397.10
April	68736.80
May	71402.75
June	68230.20
July	72557.90
August	68278.25
September	64180.05
October	64027.60
November	70395.35
December	64701.15

Recommendation

The restaurant can use these seasonal insights to plan marketing campaigns during months when revenue tends to dip. Promotional offers or limited-time menu items during slower months could help maintain steady customer traffic. Aligning staffing and inventory planning with these seasonal patterns may also improve operational efficiency and cost management.

Peak Demand Workload (Kitchen Pressure)
how many pizzas the kitchen produces per hour.
Peak Demand Workload (Kitchen Pressure)

```sql
SELECT
    HOUR(o.time) AS hour_of_day,
    ROUND(SUM(od.quantity) / COUNT(DISTINCT o.date), 2) AS avg_pizzas_per_day_at_hour
FROM orders o
JOIN order_details od
    ON o.order_id = od.order_id
GROUP BY hour_of_day
ORDER BY avg_pizzas_per_day_at_hour DESC;
```

Business Insight

The analysis shows that the kitchen experiences its highest workload during the lunch period. Around 12 PM, the kitchen produces nearly 19 pizzas per day on average during that hour, followed closely by 1 PM with around 18 pizzas. Another noticeable surge appears during the evening hours between 5 PM and 7 PM, where pizza production remains consistently high. These patterns clearly indicate that customer demand is heavily concentrated around typical meal times, especially lunch and dinner. Such demand peaks require the kitchen to handle a significantly higher production volume within a short timeframe.

hour_of_day	avg_pizzas_per_day_at_hour
12	18.98
13	17.91
18	15.17
17	14.60
19	12.34
16	11.91
14	10.35
20	10.04
15	9.06
11	8.24
21	7.46
22	5.02
9	4.00
23	2.62
10	2.25

Recommendation

The restaurant should ensure that kitchen staffing and preparation processes are optimized during these peak hours. Preparing dough, sauces, and toppings in advance can help reduce preparation time when demand spikes. Assigning specific kitchen roles during rush hours can also improve efficiency and help maintain consistent service speed.

Average Pizzas per Order by Hour (Customer Demand Intensity)

```
SELECT
    HOUR(o.time) AS hour_of_day,
    ROUND(SUM(od.quantity) / COUNT(DISTINCT o.order_id), 2) AS avg_pizzas_per_order
FROM orders o
JOIN order_details od
    ON o.order_id = od.order_id
GROUP BY hour_of_day
ORDER BY hour_of_day;
```

Business Insight

The average number of pizzas per order generally remains around two pizzas throughout most hours of the day, indicating consistent purchasing behavior among customers. During the main ordering hours, customers typically order between 2.1 and 2.7 pizzas per transaction. This pattern suggests that many orders are placed for shared meals rather than individual consumption. Early morning hours show higher averages due to a smaller number of transactions, which can slightly exaggerate the average order size.

hour_of_day	avg_pizzas_per_order
9	4.00
10	2.25
11	2.22
12	2.69
13	2.61
14	2.45
15	2.19
16	2.21
17	2.23
18	2.26
19	2.19
20	2.15
21	2.12
22	2.09
23	2.43

Recommendation

Since most customers tend to order multiple pizzas per transaction, the restaurant can benefit from offering bundle deals or group meal packages. Encouraging customers to add an additional pizza through small discounts or promotional offers could further increase the average order value. Marketing strategies can also emphasize family or group dining experiences to align with these ordering patterns.

Peak-Period Workload (Combined Peak Hours)

```sql
SELECT
    CASE
        WHEN HOUR(o.time) BETWEEN 12 AND 13 THEN 'Lunch Peak'
        WHEN HOUR(o.time) BETWEEN 17 AND 19 THEN 'Dinner Peak'
        ELSE 'Off-Peak'
    END AS period,
    ROUND(SUM(od.quantity) / COUNT(DISTINCT o.date), 2) AS avg_pizzas_per_day
FROM orders o
JOIN order_details od
    ON o.order_id = od.order_id
GROUP BY period;
```

Business Insight

When peak hours are grouped together, the dinner period shows the highest workload with an average of about 42 pizzas produced per day during those hours. The lunch period follows with roughly 37 pizzas produced on average. Off-peak hours collectively show a higher number because they cover a longer portion of the day where orders accumulate gradually. This pattern confirms that customer demand is strongly concentrated around lunch and dinner periods, creating the most operational pressure on the kitchen during these time windows.

period	avg_pizzas_per_day
Dinner Peak	41.99
Lunch Peak	36.84
Off-Peak	59.64

Recommendation

The restaurant should prioritize staffing and kitchen preparation during the lunch and dinner peaks to handle the concentrated workload efficiently. Preparing ingredients in advance and ensuring sufficient kitchen staff during these hours will help reduce delays and maintain service quality. Off-peak periods can be used for preparation work and inventory management to support smoother operations during busy times.

Top Revenue Generating Pizzas

```
SELECT
pt.name,
ROUND(SUM(od.quantity * p.price),2) AS revenue
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
JOIN pizza_types pt
ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 10;
```

Business Insight

The revenue analysis shows that the Thai Chicken Pizza generates the highest revenue, followed closely by the Barbecue Chicken and California Chicken pizzas. These items appear to be both popular among customers and priced in a way that contributes significantly to total revenue. Other strong contributors include the Classic Deluxe, Spicy Italian, and Southwest Chicken pizzas. The presence of several chicken-based pizzas among the top performers suggests that this flavor profile is particularly appealing to customers.

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768.00
The California Chicken Pizza	41409.50
The Classic Deluxe Pizza	38180.50
The Spicy Italian Pizza	34831.25
The Southwest Chicken Pizza	34705.75
The Italian Supreme Pizza	33476.75
The Hawaiian Pizza	32273.25
The Four Cheese Pizza	32265.70
The Sicilian Pizza	30940.50

Recommendation

The restaurant should highlight these high-revenue pizzas in marketing campaigns and menu placements to further increase sales. Ensuring a consistent supply of ingredients for these popular pizzas is important to avoid stock shortages during peak periods. Management could also consider creating premium variations or limited-time promotions around these successful menu items to maximize revenue potential.

Revenue by Pizza Size

```
SELECT
p.size,
ROUND(SUM(od.quantity * p.price),2) AS revenue
FROM order_details od
JOIN pizzas p
ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY revenue DESC;
```

Business Insight

Revenue distribution by pizza size shows that large pizzas generate the highest revenue, contributing approximately $375K, followed by medium and small sizes. Extra-large and double-extra-large pizzas contribute significantly less to overall revenue. This indicates that most customers prefer ordering large pizzas, likely because they provide better value for group meals or shared dining experiences. The strong performance of larger sizes aligns with the earlier finding that customers often order multiple pizzas per transaction.

size	revenue
L	375318.70
M	249382.25
S	178076.50
XL	14076.00
XXL	1006.60

Recommendation

The restaurant should focus on promoting large pizzas through combo offers or meal bundles since they already represent the largest share of revenue. Encouraging customers to upgrade from medium to large pizzas through small price incentives could further increase revenue. Promotions designed around group meals or family deals may also help capitalize on the popularity of larger pizza sizes.